SET 8: THE HOTEI REGIO CRYOVOLCANIC GEOTHERMAL TAP & HEAVY WATER EXTRACTION PLANT

DOCUMENT 8B: BORROWER'S Q3 2146 QUARTERLY REPORTING PACKAGE & VALUATION MEMO

To: Sol-System Interplanetary Development Bank (SSIDB), as Administrative Agent

From: Hotei Geothermal Energy & Isotopic Partners LP (Borrower)

Date: October 20, 2146

Subject: Q3 2146 Financial Compliance Report & Asset Valuation Analysis

1. Financial and Operational Narrative

Throughout the third quarter of 2146, the Hotei Regio facility operated in complete compliance with all covenants and technical metrics detailed in the Secured Credit Agreement (Document 8A). The cryovolcanic borehole casings maintained exceptional structural and thermal performance, posting a quarterly average **Cryovolcanic Tap Stability Rating (CTSR)** of 95.8%, which comfortably exceeds the 91.0% baseline standard specified in Section 4.01(1).

Deuterium oxide extraction volumes averaged 17,950 kilograms of pure heavy water per Titan sidereal day, successfully achieving the **Heavy Water Yield Target** of 17,500 kilograms outlined in Section 4.01(2). Consequently, the Facility Loans accrued interest at the baseline floating rate of **TIDOR plus 7.75%** per annum, with no triggers of the Thermal Runaway Containment Risk Premium. Additionally, our escrow balances in the **Localized Seismic Escrow** were fully maintained at the required SC 500,000,000 under Section 4.01(3).

2. Financial Statements (SSIAS Compliant)

Table 8.1: Consolidated Balance Sheet & Income Statement (Q3 2146)

All figures in Millions of Sol Credits (SC)

BALANCE SHEET & INCOME STATEMENT CONSOLIDATION			
Category / Account Item	**Q3 2146**	**Q2 2146**	**Variance (%)**
ASSETS			
Cash & Cash Equivalents	612.50	585.20	+4.66%
Localized Seismic Escrow[1]	500.00	500.00	+0.00%
Heavy Water & Cryogenic Gas Inventories	820.10	775.00	+5.82%
Cryovolcanic Plant & Equipment (Net)[2]	6,800.00	6,920.00	-1.73%
TOTAL ASSETS	**8,732.60**	**8,780.20**	**-0.54%**
LIABILITIES & PARTNERS' CAPITAL			
Secured Term Loans (SSIDB Facility)[3]	5,500.00	5,650.00	-2.65%
Lithosphere Restoration Provision	950.00	945.00	+0.53%
Partners' Equity Capital	2,282.60	2,185.20	+4.46%
TOTAL LIABILITIES & CAPITAL	**8,732.60**	**8,780.20**	**-0.54%**
OPERATING INCOME STATEMENT			
Gross Heavy Water & Geothermal Sales[4]	915.20	885.10	+3.40%
Borehole Thermal Ops & Centrifuge OpEx[5]	(395.10)	(391.00)	+1.05%

Net Operating Income (NOI)	**520.10**	**494.10**	**+5.26%**

[1] Reconciles directly to the mandatory localized seismic containment escrow specified in Section 4.01(3).

[2] Detailed asset class values are reconciled directly to the subtotal summation of Net Book Values reported in **Table 8.2**.

[3] Reconciles directly to the outstanding Facility Loan principal balance specified under Section 2.01.

[4] Annualized heavy water sales correlate directly with the Base Case model outlined in **Table 8.3**.

[5] Includes high-pressure valve testing, subsurface vibration monitoring, and centrifuge maintenance.

Table 8.2: Detailed Subsurface Property, Plant & Equipment (PP&E)

All figures in Millions of Sol Credits (SC)

Asset Category	Sub-Asset Class	Gross Cost	Accumulated Depr.	Net Book Value
1. Cryovolcanic Borehole	Titanium-Carbide Extraction Shaft Casing	2,800.00	(600.00)	2,200.00
	Acoustic-Resonance Monitoring Dampers	1,400.00	(320.00)	1,080.00
	Subtotal Geothermal Borehole	**4,200.00**	**(920.00)**	**3,280.00**
2. Isotopic separation	Deuterium Oxide Separation Centrifuges	1,550.00	(290.00)	1,260.00
	Cryogenic Condenser Columns	950.00	(250.00)	700.00
	Subtotal Isotopic Plant	**2,500.00**	**(540.00)**	**1,960.00**
3. Thermal Grid	High-Pressure Transmission Pipelines	1,900.00	(340.00)	1,560.00
	Subtotal Thermal Grid	**1,900.00**	**(340.00)**	**1,560.00**
TOTAL GEOTHERMAL PP&E[*]		**8,600.00**	**(1,800.00)**	**6,800.00**

[*] Reconciles directly to the **Cryovolcanic Plant & Equipment (Net)** line item on the Balance Sheet in **Table 8.1**.

3. Manager's Valuation Memo

Valuation Methodology: The Hotei Regio Cryovolcanic facility is valued utilizing a Discounted Cash Flow (DCF) model reflecting specific geothermal pressure variables and localized sub-surface safety risk parameters. Key structural assumptions include a Weighted Average Cost of Capital (WACC) of 12.50% and extraction column wear cycles over a 20-year projected operational lifespan.

Table 8.3: Net Asset Value (NAV) Modeling under Alternate Pricing Scenarios

All figures in Millions of Sol Credits (SC) unless specified

Valuation Parameter	Base Case (Heavy Water @ SC 520/kg)	Bear Case (Heavy Water @ SC 390/kg)	Bull Case (Heavy Water @ SC 650/kg)
Annual Heavy Water Output (M kg)	7.04	6.50	7.50
Gross Revenue (SC/yr)[#]	3,660.80	2,535.00	4,875.00
Borehole Operations & Utility Costs	(1,580.40)	(1,620.00)	(1,540.00)
Acoustic-Resonance Recalibrations	(500.00)	(500.00)	(500.00)
Net Operating Income (NOI)	**1,580.40**	**415.00**	**2,835.00**
Capitalized Value (8.5% Capitalization Rate)	**18,592.94**	**4,882.35**	**33,352.94**
Debt Outstanding (SSIDB Facility)[^]	(5,500.00)	(5,500.00)	(5,500.00)
Implied Net Asset Value (NAV)	**13,092.94**	**(617.65)**	**27,852.94**

[#] Pro-rata Q3 Gross Revenue matches the **Gross Heavy Water & Geothermal Sales** reported in **Table 8.1** (SC 915.20m represents exactly 25.0% of the annualized Base Case Gross Revenue of SC 3,660.80m).
[^] Corresponds to the **Secured Term Loans (SSIDB Facility)** balance reported in the liabilities section of **Table 8.1**.

4. Five-Year Discounted Cash Flow (DCF) Schedule

Projection Summary: Below is the detailed 5-year financial model for Hotei Geothermal. It incorporates inflation-linked operation cost growth of 3.0% annually, a discount rate of 12.50%, and an exit capitalization rate of 8.5% applied to Year 5 Free Cash Flow.

Table 8.4: Detailed Five-Year Free Cash Flow Projection

All figures in Millions of Sol Credits (SC) unless specified

Financial Parameter	Year 1	Year 2	Year 3	Year 4	Year 5
Estimated Heavy Water Yield (M kg)	7.04	7.25	7.47	7.69	7.92
Gross Sales Revenue	3,660.80	3,770.00	3,884.40	4,001.00	4,118.40
Operating and Utility Expenses	(1,580.40)	(1,627.80)	(1,676.60)	(1,726.90)	(1,778.70)
Acoustic-Resonance Audits	(500.00)	(515.00)	(530.50)	(546.40)	(562.80)
Net Operating Income (NOI)	**1,580.40**	**1,627.20**	**1,677.30**	**1,727.70**	**1,776.90**
Less: Maintenance Capital Expenditures	(250.00)	(257.50)	(265.20)	(273.20)	(281.40)
Free Cash Flow to Firm (FCFF)	**1,330.40**	**1,369.70**	**1,412.10**	**1,454.50**	**1,495.50**
Discount Factor (at 12.50%)	0.8889	0.7901	0.7023	0.6243	0.5549
Present Value (PV) of FCFF	**1,182.59**	**1,082.20**	**991.72**	**908.04**	**829.85**
Terminal Value (exit capitalization @ 8.5%)	—	—	—	—	17,594.12
Present Value of Terminal Value	**—**	**—**	**—**	**—**	**9,762.98**

[*] Cumulative Net Enterprise Value sum of Year 1-5 cash flows and present value of terminal value translates directly to **SC 14,757.38 million**. Reconciling this against outstanding debt yields robust mezzanine and equity value coverage under the specified parameters.